

Lab Assembly Final T1.2023-2024
(30 Marks – Theoretical)

Question #1 (Multiple Choice Questions)

Q1) Equivalent Instruction of STOSW , DF = 1 :

Answer : MOV [DI],AX ; DEC DI ; DEC DI ;

Q2) In the Following Instruction, Value of AL in Binary :

MOV AL,FF_H

NEG AL

XOR AL,FF_H

Answer : AL = 11111110_B

Q3) The Value of the AX Register after Executing the Following Code, if The Value of SP = FFFA_H :

DEC SP

INC SP

INC SP

PUSH SP

MOV AX,SP

Answer : SP = FFF9_H

Q4) Find the Content of AL and ZF after Executing the following Code :

MOV AL,58_H

AND AL,4

Answer : AL = 0 , ZF = 1

Q5) The Value of AL After Input Number 3 from KBD and Multiply it with 2 :

Answer : 66_H

Q6) Which of Following Code Used to Convert 7A_H to FB_H :

Answer : OR BX,81_H

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Lab Assembly Final T1.2023-2024
(30 Marks – Theoretical)

Q7) Which of Following Instruction Change CF Value :

Answer : SHL , SHR , ROR and RCR

Q8) If DF = 0 (_____) Instruction Increment SI Register One Every Time :

Answer : LODSB

Q9) Instruction (_____) Will Divide The Unsigned Number By 2 On Each Time :

Answer : SHR

Q10) Find the Content of BL Register After Executing the Following Code :

MOV BL,0F_H

SAL BL,1

ADC BL,1

Answer : 1F_H

Lab Assembly Final T1.2023-2024
(30 Marks – Theoretical)

Question #2 (Trace)

Q1) `MOV AX,02FFH`

`MOV CL,0FCH`

`ADD AL,5`

`SUB AL,0`

`XCHG AL,AH`

`MUL CL`

`AX = (01F8H) , CL = (FCH)`

Q2) `X DB 16,4,32H`

`MOV AL,X[0]`

`MOV BL,X[2]`

`X[0] = (10H) , X[2] = (32H)`

Q3) The Value of Variable Y :

`X DB 'ASASASASA'`

`Y DB 5 DUP('N')`

`CLD`

`MOV SI,OFFSET X`

`MOV DI,OFFSET Y`

`MOV CX,3`

`REP MOVSB`

`MOV AL,'#'`

`STOSB`

`Y = ("ASA#N")`

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Lab Assembly Final T1.2023-2024
(30 Marks – Theoretical)

Q4) *ARRAY DB 7,7,8,8,7,7,8,8,8,8*

MOV AL,0

MOV CX,4

MOV BX,OFFSET ARRAY

L:STC

ADC AL,[BX]

LOOP L

MOV AH,[BX]

AL = (20_H) , AH = (07_H) , CF = (0)

Q5) *Value Y :*

MOV DX,0

MOV CX,3

MOV SI,0

STC

L:

SBB DL,X[SI]

INC SI

LOOP L

MOV Y,DL

X DB 3,-8,4

Y DB ?

Y = (FF_H)

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Lab Assembly Final T1.2023-2024
(30 Marks – Theoretical)

Q6) Suppose the User Enter :

ABD\$#qE

What's the Output :

L1: MOV AH,01H

INT 21H

CMP AL,'#'

JE L2

MOV DL,AL

MOV AH,02H

INT 21H

JMP L1

L2:

Output : AABDD\$#\$

Lab Assembly Final T1.2023-2024
(30 Marks – Theoretical)

Question #3 (Trace INT Code)

Q1) Trace the Following Code And Show what Results will Appear on Screen and/or any other Variable , Note ; Tab = 8 Spaces , '\$' = 24H

STR DB 35H,30H,33H,24H

MOV AH,09H

LEA DX,STR

INT 21H

MOV BX,DX

ADD BX,3

MOV DL,[BX]

MOV AH,02H

INT 21H

Solution:

5	0	3	\$						

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